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Paper

Lightweight Object Detection Model for a CMOS Image Sensor with Binary Feature Extraction

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Abstract Anticipating the rise of the Internet of Things (IoT) era, we have proposed an object detection framework that employs a CMOS image sensor with binary feature extraction to reduce power requirements. Initially, we presented a lightweight deep neural network for the feature data based on the YOLOv7, comparable to the YOLOv7-tiny in the number of parameters and FLOPs, but it enhances large object recognition accuracy (APL50) by 6.6%. Moreover, our approach achieves a 48.8% reduction of GPU power consumption compared to the YOLOv7. Additionally, we introduce an on-chip signal processing method for the binary feature data. The proposed method achieves a compression rate of 64.1% and increases GPU power consumption by only 14.9% during the decoding process preceding object detection. Moreover, the size of 1-bit feature data is reduced by 96.0%, and object recognition accuracy is improved by 4.0% relative to 1-bit RGB color images.

Key words: CMOS image sensor, feature extraction, object detection, data reduction, run-length encoding

1. Introduction

In the present Internet of Things (IoT) era, big data from numerous physical sensors is transmitted to cyberspace[1], where artificial intelligence (AI) processes it and provides useful feedback to the physical space. For sustainability purposes, it is critical to reduce the power consumption of a recognition system composed of sensors and AI. In the realm of image sensing, CMOS image sensors for AI applications, known as sensing image sensors, have emerged as a promising solution.

Figure 1 shows a comparison between viewing image sensors and sensing image sensors in an image recognition system. Viewing image sensors generate high-resolution photographic images intended for human observation; however, these images typically contain redundant information that is later pruned by a deep neural network (DNN) during feature extraction. In contrast, sensing image sensors directly produce lightweight feature data suitable for AI tasks, thereby lowering both the computational complexity of the DNN and the storage requirements for vast imaging data[3)4)5)]. For instance, a log-gradient QVGA image sensor[3)] outputs feature data in the form of histograms of oriented gradients (HOGs). However, its readout circuitry is specialized for HOG extraction, which prevents it from generating standard photographic images. A convolutional image sensor[4)] is capable of outputting both photographic images and feature data. However, they require in-pixel capacitors to perform multiply–accumulate (MAC) operations, resulting in larger pixels that may adversely affect light sensitivity and increase chip costs. Event-based vision sensors (EVSs)[5)] are designed to suppress temporal redundancy and support high dynamic range operations with lightweight feature data. However, EVSs struggle to capture stationary objects because they omit small luminance variations (those falling below and rising above a set threshold[6)7)]) to avoid redundancy. Moreover, their pixel sizes are increased due to the required in-pixel circuits for de-

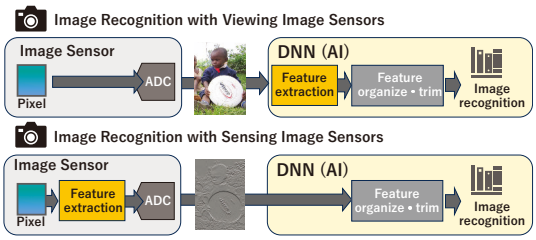


Fig. 1 The difference between image recognition systems using viewing image sensors and sensing image sensors[2)].

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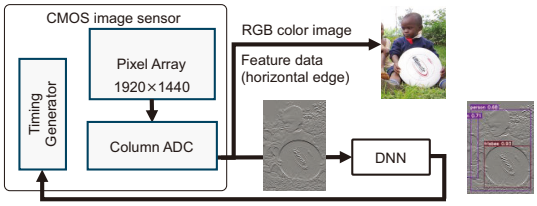


Fig. 2 Object recognition system with a feature-extractable CMOS image sensor[2].

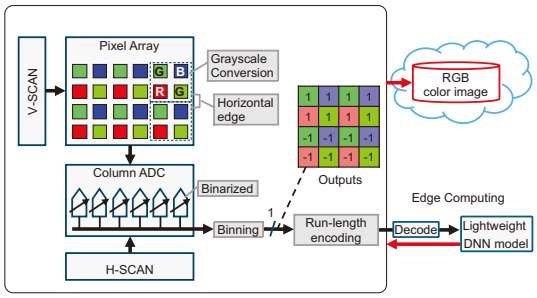


Fig. 3 Proposed object detection system with a feature-extractable CMOS image sensor.

tecting temporal changes.

Our research group has proposed a CMOS image sensor capable of producing both photographic RGB color images and feature data[8)9)]. This feature-extractable CMOS image sensor is designed to output simple horizontal edge data while maintaining compatibility with standard RGB color image sensors that use small pixels.

Figure 2 shows the conceptual framework of an object recognition system that incorporates the image sensor with a 1920×1440 Bayer pixel array. By utilizing lightweight feature data processed by a lightweight DNN model deployed at the edge, the overall power consumption of the recognition system is significantly reduced by eliminating the need to transfer big image data to cyberspace. When a given object is detected in the feature data, the sensor switches its operating mode to capture RGB color images for human viewing or for further analysis by a DNN in cyberspace.

Previously, our group proposed an on-chip signal processing method for CMOS image sensors that extracts 3-bit feature data[9)]. We evaluated the accuracy of the feature data in object recognition using YOLOX[10)], which was designed for a normal GPU and trained with a simulated feature dataset. Our simulation results showed that the amount of 3-bit feature data was reduced by 97.7% compared to conventional RGB color images, while the decrease in object recognition accuracy was limited to only 13.0% (= 69.6%−56.6%), as evaluated on the COCO dataset[2)] In this paper, we present a novel lightweight DNN model based on YOLOv7[11)] that processed the feature data with a low power consumption, as described in Section 2. This model can be implemented on the NVIDIA's edge AI device, Jetson Nano. Furthermore, we propose an on-chip signal processing technique for the CMOS image sensor to extract binary feature data using run-length encoding for efficient signal compression, as described in Section 3. Section 4 summarizes this paper.

The main contributions of this study are summarized as follows:

- We present a lightweight DNN model based on YOLOv7 that processes the feature data with a low power consumption.
- We introduce an on-chip signal processing method for CMOS image sensors that extracts binary feature data through run-length encoding, allowing efficient signal compression.

Update from the Conference Proceeding

A conference proceeding version of this paper appeared in the IEEE SENSORS 2024[12)], and this extended version contains the following new contents:

- A comparison of our methods with other approaches (Section 1);
- Details of the ELAN modules used for evaluation (Figure 5);
- Simulation results on the PASCAL VOC dataset (Section 2. 4);
- GPU power consumption measurements on the Jetson Nano for object detection (Section 2. 5);
- An explanation of the data flow of run-length encoding for the binary feature data (Figure 10);
- Power consumption measurement for decoding binary feature data with run-length encoding prior to input the lightweight DNN (Section 3. 3);
- Sample images of recognition results (Figure 13).

2. Lightweight DNN Model for Feature Data

Figure 3 shows a more detailed depiction of the system illustrated in Figure 2. The RGB color image is read out using a pixel-by-pixel architecture at an 8-bit resolution, similar to viewing CMOS image sensors. Feature data are extracted as follows. First, using four-

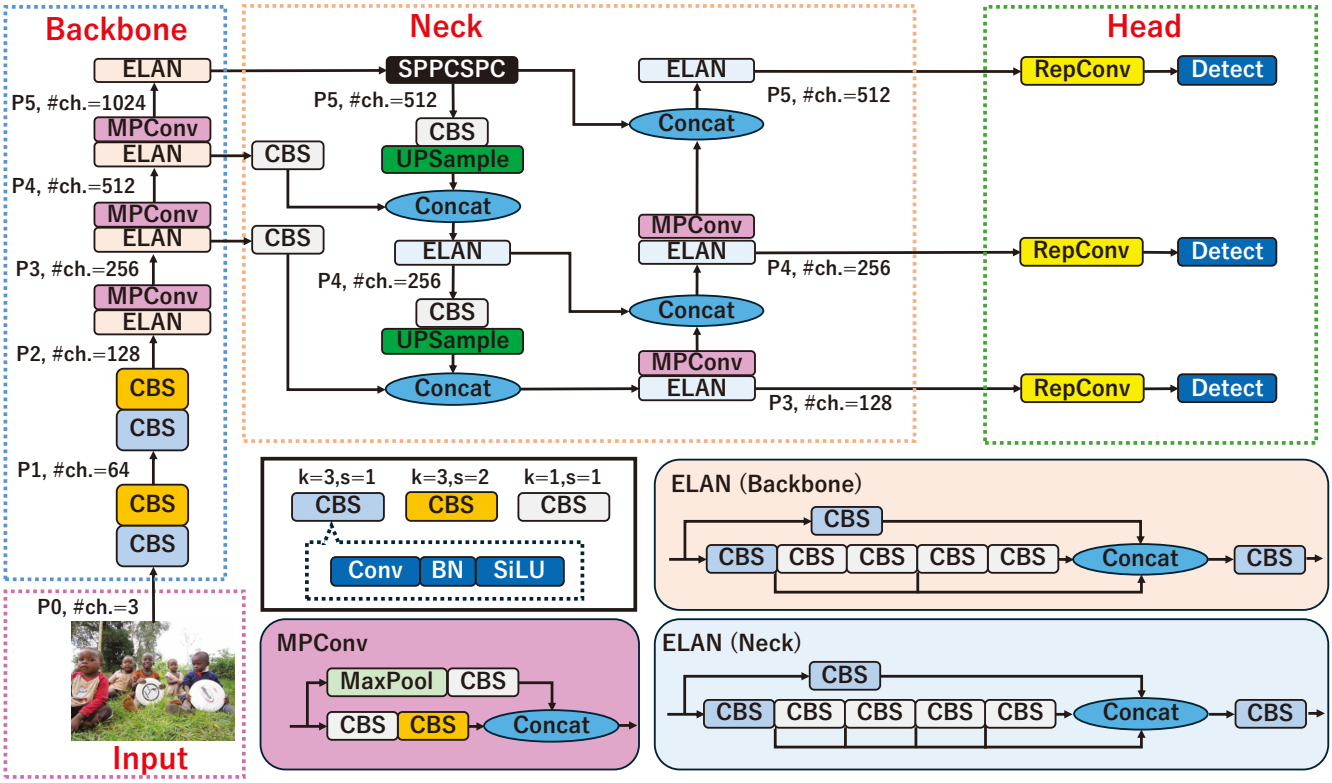


Fig. 4 YOLOv7 network architecture[11]. The sizes of feature maps are denoted as P{resolution level}, and #ch. represents width. The valuables of k and s means kernel size and stride of the convolutional layer, respectively.

shared pixel structure, the sensor converts the RGB pixel signal into a grayscale signal using the formula $R+2G+B$[13]. This simultaneous readout of photoelectrons from the Bayer array reduces the spatial resolution to one-fourth. Next, instead of performing correlated double-sampling on a single pixel, the horizontal edge is extracted by subtracting the signals from vertically adjacent Bayer cells, which further halves the spatial resolution. The analog feature output from the pixel array is then quantized to a 1-bit signal via a column-parallel ADC. Subsequently, the horizontally adjacent pixel signals are binned together, causing another halving of the spatial resolution. Finally, run-length encoding is applied to the extracted binary feature data. These processing steps compress the feature output from the sensor, thereby lowering its power consumption[8]. The encoded binary feature data are subsequently decoded and processed by the lightweight DNN model based on YOLOv7[11].

2.1 YOLOv7 Architecture and Feature Adaptation

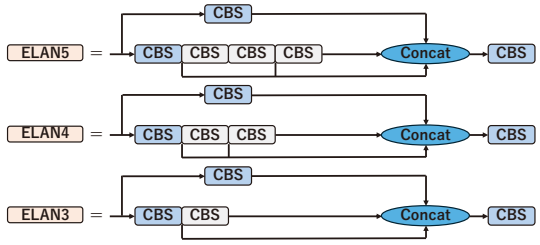
YOLOv7 is an object detection model that introduces well-tuned network computation modules and training techniques to improve recognition accuracy without increasing inference costs. YOLOv7-tiny[11] is a compact and efficient version of YOLOv7 and tailored for edge GPUs. In this section, we similarly make YOLOv7 compact and lightweight by reducing both width (the number of channels) and its depth (the number of stacked layers), focusing on 8-bit feature data.

YOLOv7 network architecture consists of four components: input, backbone, neck and head, as shown in Figure 4. As the network depth increases, convolutional layers and downsampling operations are applied repeatedly, reducing the spatial resolution while increasing width. The main network modules of YOLOv7 are the ELAN modules, which play a significant role in feature extraction. ELAN modules in the backbone and neck consists of six stacked CBS modules, which is composed of convolution (Conv), batch normalization (BN), and SiLU activation functions, with concatenation-based skip connections.

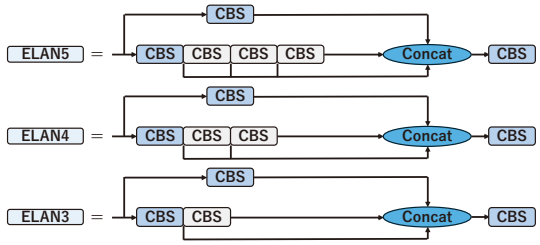
To simulate object recognition accuracy with a feature-extractable CMOS image sensor, we convert the RGB color images from the COCO dataset into the feature dataset[9]. The feature dataset contains 30,000 training images, and 5,000 validation images. All models we mentioned are trained from scratch without using pre-trained weights.

Table 1 Object recognition accuracy for width reduction

Model	Width	#Param.	FLOPs	AP50	AP_L50
YOLOv7	100%	36.9M	58.5G	52.2%	69.6%
x0.75	75%	20.9M	33.1G	50.2%	68.1%
x0.5	50%	9.4M	14.9G	46.5%	63.7%
x0.25	25%	2.4M	3.8G	36.9%	55.4%
YOLOv7-tiny	50%	6.2M	7.6G	37.7%	55.8%



(a) Backbone.



(b) Neck.

Fig. 5 ELAN modules used for evaluation.

2.2 Width Reduction

Width of the backbone and neck in the network is reduced to 0.75, 0.5, and 0.25 times that of original YOLOv7. Table 1 shows the comparison of the width reduction models and YOLOv7-tiny for reference. Object recognition accuracy is quantified using AP50 (mean average precision at an IoU of 0.50) and AP_L50 (accuracy for large objects only). The reason we focus on large objects is that the proposed feature-extractable CMOS image sensor assumes that if given object is detected in the feature data, then further analysis, such as detecting small objects, being performed on the RGB color images.

Since the degradation in object recognition accuracy is significant at 0.25 times width, we chose a 0.5 reduction factor for width.

2.3 Depth Reduction

In addition to width reduction, depth in the network is also reduced. Depth reduction is applied to the ELAN modules.

Figure 5 shows the ELAN modules in the backbone and neck used for evaluation. ELAN5, ELAN4, and ELAN3 consist of five, four, and three-stacked CBS modules, respectively. However, since reducing depth simultaneously decreases width of the concatenated feature maps by the Concat modules, the compound scaling method[11)14)] is applied to the depth reduction models. Table 2 shows the comparison of the depth reduction models and YOLOv7-tiny, where D_B and D_N represent ELAN depth of the backbone and neck, respectively. Although reducing depth in the backbone (D_B) leads to a notable decline in object recognition accuracy, similar reductions in the neck (D_N) have a smaller adverse effect. In fact, the model with $D_N = 3$ achieves a smaller number of parameters compared to YOLOv7-tiny, while improving AP50 and AP_L50 by 7.2% and 6.6%, respectively.

As a result, the lightweight model with 0.5 times of width and the ELAN3 module in the neck is presented for the feature data.

2.4 Evaluation on Another Dataset

To examine the robustness of the proposed model under different conditions, we additionally evaluated the object recognition accuracy using the PASCAL VOC dataset[15)]. The PASCAL VOC dataset was converted into the 8-bit feature data format in the same manner as the COCO dataset. Table 3 shows the object recognition accuracy on the PASCAL VOC dataset, comparing the presented lightweight model with the YOLOv7-tiny. The lightweight model achieves a smaller number of parameters compared to the YOLOv7-tiny, while improving AP50 by 6.3%, similar to the improvement on the COCO dataset. These results demonstrate that the proposed model exhibits robustness for different datasets. In the following simulations, we use the COCO dataset.

2.5 Power Consumption

To assess the power efficiency of the presented lightweight model, we measured the power consumption of RTX 3060 GPU with 3,000 test feature data. As shown in Fig. 6, the power consumption of the lightweight model and YOLOv7-tiny were 48.8% and 60.9% lower, respectively, compared to the original YOLOv7. The power consumption trend of the proposed model and YOLOv7-tiny correlates with the FLOPs, which are 11.8 G and 7.8 G, respectively. We also measured GPU power consumption on the NVIDIA's edge AI device, Jetson Nano, using 1,000 test feature data. On this platform, the presented lightweight model can be implemented on the Jetson Nano, similarly to YOLOv7-tiny, while the original YOLOv7 cannot be implemented due to large model size and memory constraints. The lightweight model consumed 326 mJ per image, while YOLOv7-tiny con-

Table 2 Object recognition accuracy for depth reduction

Model	Width	D_B	D_N	#Param.	FLOPs	AP50	AP_L50
This work	50%	6	6	9.4M	14.9G	46.5%	63.7%
	50%	5	6	9.0M	14.0G	45.4%	63.4%
	50%	4	6	8.7M	13.1G	45.2%	62.5%
	50%	3	6	7.4M	10.7G	42.2%	59.6%
	50%	6	5	8.1M	13.7G	46.0%	63.3%
	50%	6	4	7.0M	12.7G	44.1%	61.6%
	50%	**6**	**3**	**5.9M**	**11.8G**	**44.9%**	**62.4%**
YOLOv7-tiny	50%	4	4	6.2M	7.6G	37.7%	55.8%

[1] D_B indicates depth in the backbone ELAN.
[2] D_N indicates depth in the neck ELAN.

Table 3 Object recognition accuracy on PASCAL VOC dataset[15)]

Model	#Param.	FLOPs	AP50
Lightweight model	5.9M	11.8G	80.4%
YOLOv7-tiny	6.2M	7.6G	74.1%

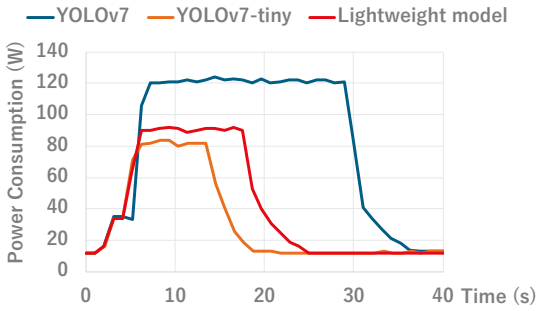
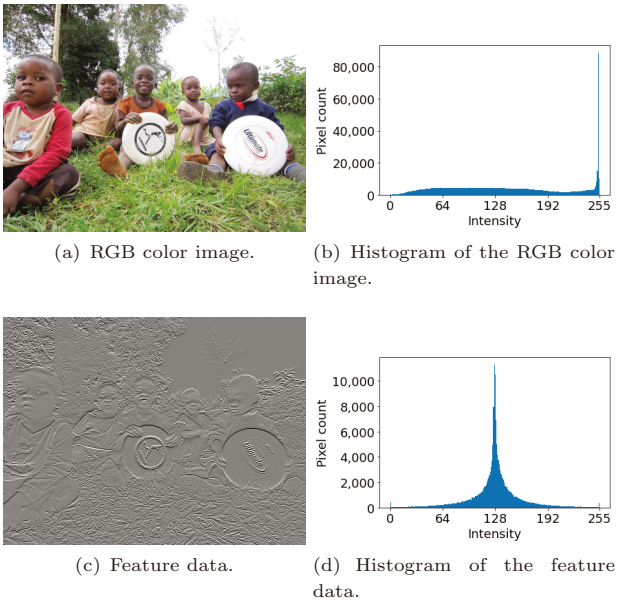


Fig. 6 Power consumption of YOLOv7, YOLOv7-tiny and the presented lightweight model.



(a) RGB color image. (b) Histogram of the RGB color image.

(c) Feature data. (d) Histogram of the feature data.

Fig. 7 Sample images from the COCO dataset[2)] and their respective histograms.

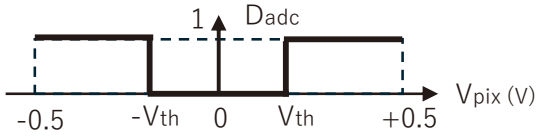
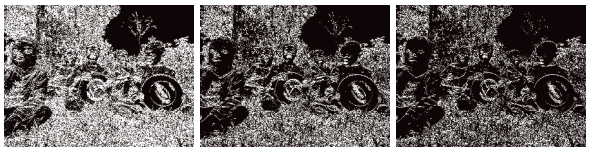


Fig. 8 Overview of quantization in a CMOS image sensor.



(a) V_{th} = 7.5 mV (b) V_{th} = 15 mV (c) V_{th} = 22.5 mV

Fig. 9 Binary feature data generated by the COCO dataset[2)].

sumed 204 mJ per image. Although the proposed lightweight model consumes slightly more power than YOLOv7-tiny, it is noted that the proposed model is sufficiently compact for implementation on edge AI platforms, and achieves over 5% higher object recognition accuracy.

3. Binary Feature-Extractable CMOS Image Sensor

In this section, we propose an on-chip signal compression method for the feature data.

3.1 Binarization Technique of Feature Data

Figure 7 shows examples comparing a 8-bit RGB color image with its corresponding feature data. The histogram distributions of Figs. 7(a) and (c) are described in Figs. 7(b) and (d), respectively. In the feature data, the pixel intensities above 128 denote a positive gradient, those below 128 indicate a negative gradient, and an intensity of exactly 128 means no gradient. However, the intensity of non-gradient pixels distributes around 128 due to random noise. Accordingly, we propose a binarization technique in the column ADC where the analog feature data (V_{pix}) is converted to a binary value: it becomes 1 if it is above or below a predetermined threshold voltage (V_{th}), and 0 otherwise, as shown in Fig. 8. Fig. 9 shows binary feature outputs for V_{th} values of 7.5 mV, 15 mV, and 22.5 mV, respectively. A lower threshold preserves much redundant edge information, while a higher threshold results in the loss of significant edge details.

Table 4 summarizes the impact of different V_{th} set-

Table 4 The impact of V_{th} settings on object recognition accuracy

V_{th}	AP50	AP_L50
7.5 mV	26.4%	44.2%
15 mV	27.6%	44.7%
22.5 mV	27.2%	43.7%

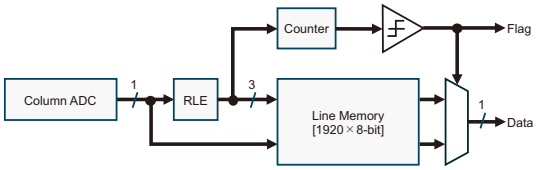


Fig. 10 The data flow diagram of run-length encoding for the binary feature data.

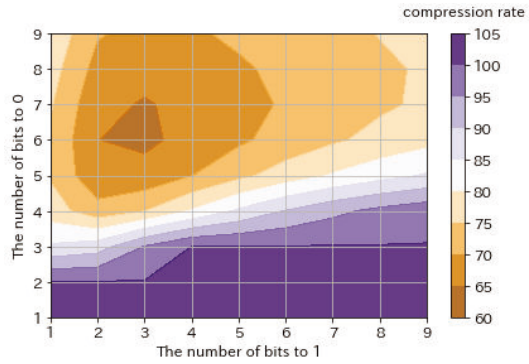


Fig. 11 The compression rate based on bit allocation.

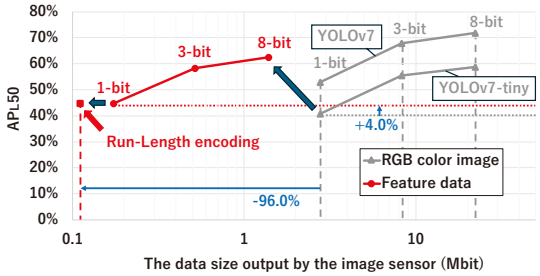


Fig. 12 Object recognition accuracy and data size.

tings on object recognition accuracy when the feature dataset is quantized to 1-bit using our proposed method. The highest AP50 and AP_L50 is achieved at V_{th} = 15 mV, which is therefore adopted for the binarization process.

3.2 Run-Length Encoding for Binary Feature Data

Run-length encoding compresses binary feature data by counting successive occurrences of zeros and ones. For instance, given the sentence "0000,0111", the five consecutive zeros might be encoded as "101" and the three consecutive ones as "11". Since the binarized feature data is composed of only zeros and ones, we propose a run-length encoding scheme that alternates the count of consecutive zeros and ones within a given bit-length. Thus, the sentence "0000,0111" is encoded to "101,011" when 3-bit is allocated for zeros and ones. Overflows occur when the number of consecutive zeros or ones exceeds the capacity of the counter bits. For example, with a 3-bit counter, the sequence "0000,0000" is encoded as "111,000,001", where the count is divided into multiple segments to accommodate the overflow. This scheme ensures that the original data can be fully reconstructed, although it reduces compression efficiency.

Figure 10 shows the data flow diagram of run-length encoding for the binary feature data. The feature extractable CMOS image sensor employs a 1920×8-bit line memory for RGB color images. The 480×1-bit ADC output is written to the line memory, and is also encoded using the proposed run-length scheme. It is noted that the one frame feature data size is 480×320 pixels derived from a 1920×1440 Bayer pixel array. For example, when 3-bit is allocated for zeros and ones, the successive length of zeros and ones are alternatively written to the free space of the line memory. The length of the encoded sentence is also counted simultaneously. If the counted length is less than 480, the compression rate is less than 100% and the encoded data is output along with a flag for proper decoding. Otherwise, if the compression rate will be above 100%, for example due to the presence of many overflows, the 480×1-bit ADC output in the line memory without encoding is selected to keep the compression rate less than 100%.

Figure 11 shows the compression rates achieved when our method is applied to the binary feature data from 25,000 test images. We found that allocating 6-bits for zeros and 3-bits for ones yields a maximum compression rate of 64.1%. This outcome is due to the fact that the absence of an edge gradient (denoted by zeros) requires a longer bit-length compared to the presence of an edge gradient (denoted by ones).

3.3 Decoding Power Consumption

In addition to evaluating compression efficiency, we measured the GPU power consumption on Jetson Nano during the decoding compressed binary feature data with proposed run-length encoding scheme. The decoding process consumed 48.4 mJ per image, accounting for 14.9% of the GPU power consumption for object detection using the presented lightweight model.

3.4 Object Recognition Accuracy and Data Size

Figure 12 shows the object recognition accuracy (AP_L50) and data sizes for a YOLOv7-tiny model

Table 5 Object recognition accuracy and data size in quantized RGB color images.

Bit depth	8-bit	3-bit	1-bit
Data size (bit)	22,118,400	8,294,400	2,764,800
AP50	48.1%	45.3%	31.6%
AP_L50	58.7%	55.5%	40.7%

Table 6 Object recognition accuracy and data size in quantized feature data.

Bit depth	8-bit	3-bit	1-bit	1-bit +Compression
Data size (bit)	1,382,400	518,400	172,800	110,765
AP50	44.9%	39.8%	27.6%	
AP_L50	62.4%	58.4%	44.7%	



(a) The RGB color image. (b) The binary feature data.

Fig. 13 Sample images of object recognition results[2].

trained on quantized RGB color images and the presented lightweight model trained on quantized feature data. For reference, the accuracy of YOLOv7 model is also plotted. Detailed numerical results are provided in Tables 5 and 6. The results indicate that feature data yield superior object recognition accuracy compared to RGB color images, despite having smaller data sizes. The size of the feature data with the run-length encoding is reduced by 96.0%, and object recognition accuracy is improved by 4.0% compared to 1-bit RGB color images. When we define efficiency as the ratio AP_L50/(Data Size), the presented lightweight model operating on binary feature data achieves an efficiency of 403.6 %/Mbit; that is 27.4 times greater than that of the original YOLOv7-tiny using 1-bit RGB color images.

Figure 13 shows sample recognition results focused on large objects; although the confidence scores are somewhat lower, all objects are correctly detected using our approach.

4. Conclusions

We have proposed an object detection system using a CMOS image sensor capable of extracting binary feature data. The presented lightweight model for processing 8-bit feature data improved AP_L50 by 6.6%, despite having fewer parameters than YOLOv7-tiny. Additionally, this model reduced GPU power consumption by 48.8% compared to the original YOLOv7. An on-chip signal processing technique for the CMOS image sensor was also presented to extract compressed binary feature data. The simulation results show that our approach reduces the feature data size by 96.0%, and object recognition accuracy is improved by 4.0% relative to 1-bit RGB color images. Overall, the efficiency of the presented DNN using feature data is 27.4 times higher than that of the original YOLOv7-tiny with 1-bit RGB color images. Our future work will focus on implementing this method in hardware to develop a low-power image recognition system.

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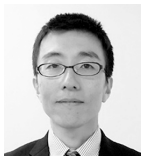
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